Data Driven Modeling of Transient Hysteresis Characteristics of Power Magnetics under Non Sinusoidal Excitations

Transformer Model

Haswika C, Dr. B. Sai Ram and Dr. A. M. Tripathi
Indian Institute Of Technology Dharwad, Department Electric Engineering

Abstract

A data-driven framework is proposed to model magnetic hysteresis behavior under non-sinusoidal excitations. The method learns the nonlinear mapping between magnetic flux density $B(t)$ and magnetic field intensity $H(t)$ using a Transformer-based sequence learning approach.

To improve generalization across operating conditions, a knowledge distillation strategy is employed, where multiple frequency-specific teacher models guide a unified student model.

Introduction

- Magnetic hysteresis in power electronic systems exhibits strong nonlinear and history-dependent behavior, especially under non-sinusoidal excitations such as PWM waveforms.
- Classical analytical approaches are often limited to sinusoidal steady-state conditions. As a result, they fail to capture transient dynamics and frequency-dependent behavior observed in real operating conditions.

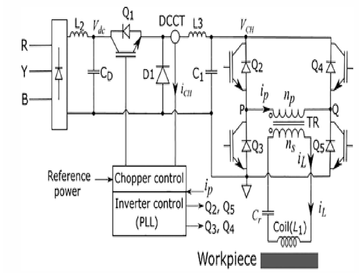


Figure 1: A resonant converter for induction heating

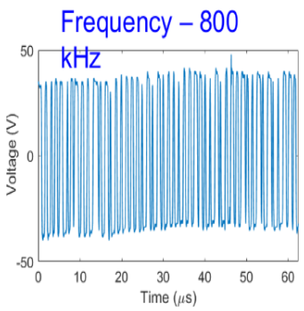
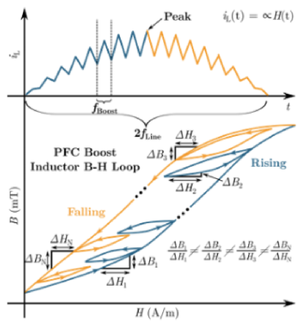


Figure 2: Voltage form excitations



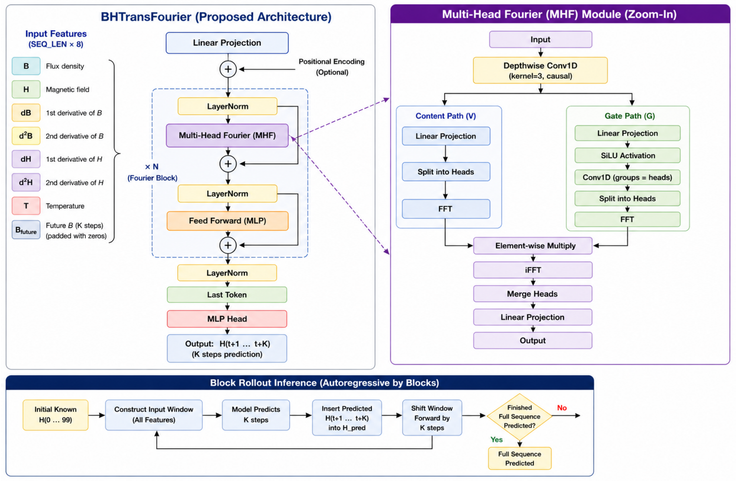


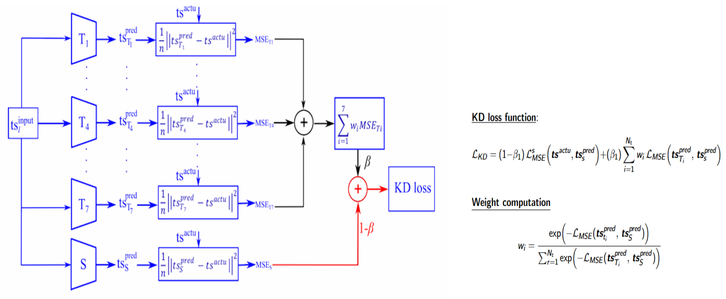
- With the increasing complexity of modern power systems, there is a growing need for data-driven approaches that can learn temporal dependencies directly from measurements.
- Sequence modeling techniques, particularly Transformer-based architectures, provide a powerful framework for capturing long-range dependencies in time-series data.

Proposed Methodology

Model Architecture:

- Input sequence is passed through a linear embedding layer
- Transformer encoder learns long-range dependencies
- Multi-head attention captures nonlinear relationships





KD loss function:

$$\mathcal{L}_{KD} = (1-\beta_1)\,\mathcal{L}^{s}_{MSE}\left(ts^{actu}, ts^{pred}_{S}\right) + (\beta_1)\sum_{i=1}^{N_t} w_i\,\mathcal{L}_{MSE}\left(ts^{pred}_{T_i}, ts^{pred}_{S}\right)$$

Weight computation

$$w_i = \frac{\exp\left(-\mathcal{L}_{MSE}\left(ts^{pred}_{T_i}, ts^{pred}_{S}\right)\right)}{\sum_{i=1}^{N_t}\exp\left(-\mathcal{L}_{MSE}\left(ts^{pred}_{T_i}, ts^{pred}_{S}\right)\right)}$$

Figure 3: A resonant converter for induction heating

Dataset & Task

- Inputs: time-series $B(t)$, $H(t)$, $T(t)$ with frequency labels (1–7)
- Training set: 15 sequences (limited labeled data)
- Test set: 1260 sequences (no frequency labels)
- Given first 100 time steps → predict next 900 steps of $H(t)$

Key Challenge: Long-horizon prediction with limited labeled data and unlabeled testing conditions.

Performance of the model

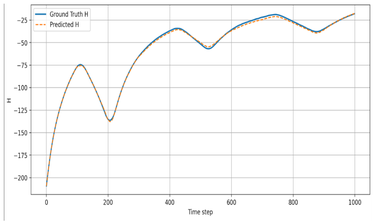
Overall Error

Method	Errseq	Errene
Baseline Model	41.03	4.37
KD Method	36.05	2.18

Frequency-wise Performance

Freq (kHz)	Baseline		KD	
	Seq	Ene	Seq	Ene
50	54	2.7	55	1.2
80	54	3.4	56	1.5
125	37	4.1	39	1.6
200	26	4.2	25	1.8
320	29	4.2	22	1.7
500	33	3.7	23	1.5
800	46	5.5	18	3.0

Best Results



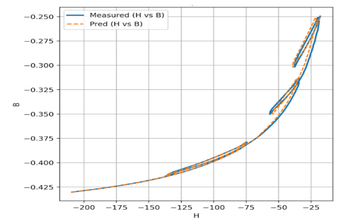
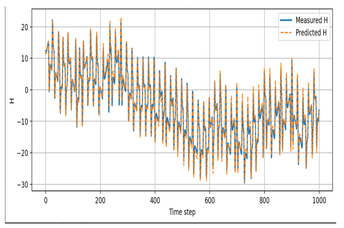
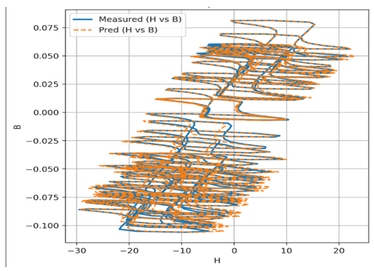


Figure 4: H predicted

Figure 5: BH loop





Conclusion

- A Transformer-based data-driven model successfully captures nonlinear and history-dependent hysteresis behavior.
- Knowledge Distillation improves generalization across frequencies and reduces prediction error.
- The approach is effective even with limited labeled training data.

References

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[2] G. Bertotti, *Hysteresis in Magnetism*, Academic Press, 1998.
[3] Hinton et al., "Distilling the Knowledge in a Neural Network," 2015.
[4] MAGnet Challenge 2, IEEE PELS Data-Driven Magnetics Modeling Competition